RiskGNN++: classification, evidence learning, and selective review

A PRIMARY PREDICTION

Shared backbone
Intrinsic + graph + hypergraph

→ **Learned representation mix**
Node and graph representations

→ **Primary softmax head**
Discriminative prediction

→ **Probability calibration**
UK: isotonic / SMEsD: Platt

calibrated p

B EVIDENTIAL OPINIONS AND REVIEW

Evidential heads
Node + relation-conditioned

→ **Opinion bundle**
Node and relation opinions

→ **Composite conflict C**
Computed before discounting

C →

Review score
ambiguity(p) + gamma * C
Defer the highest scores

Relation pooling includes self;
missing relations still yield opinions.

Shared representations:
opinions are not independent sources.

Evaluate at matched review budgets

opinions + degree

C AUXILIARY FUSION

Training coupling
Auxiliary losses update
shared model parameters.

Conflict / degree discount
Applied to relation opinions

→ **Dempster fusion**
Then degree-based scaling

→ **Auxiliary objective**
Fused evidence + KL regularization

→ Primary probabilities
→ Opinion / conflict features
→ Auxiliary evidence pathway

Implemented paths. The primary head does not consume fused probabilities; scalar fused uncertainty is a separate review baseline.